BOARDROOM-READY MATH · 2026

Math Layouts

Seven variants under one `math` class. For quants, mathematicians, physicists, ML researchers, statisticians, and economists. KaTeX renders `$...$` inline and `$$...$$` display; the layouts surround the equations with the structure each audience expects.

SECTION 01

The hero equation · bare `math` resolves to feature.

The closed-form estimator.

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

WHERE

- $\hat{\beta}$ — OLS coefficient vector
- X — design matrix, $n \times p$
- y — response vector, length n
- $X^\top X$ — Gram matrix, $p \times p$, must be invertible

SECTION 02

The derivation chain · `math derivation`

Derivative of f from first principles.

$f(x+h) = f(x) + f'(x)\,h + O(h^2)$	*Taylor expansion,* $n = 2$
$f(x+h) - f(x) = f'(x)\,h + O(h^2)$	*subtract* $f(x)$
$\dfrac{f(x+h) - f(x)}{h} = f'(x) + O(h)$	*divide by* $h \neq 0$
$\lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h} = f'(x)$	**take the limit**

SECTION 03

Definition · theorem · proof · `math theorem`

Intermediate Value Theorem.

DEFINITION. A function $f : [a,b] \to \mathbb{R}$ is *continuous* on $[a,b]$ if $\lim_{x\to c} f(x) = f(c)$ for every $c \in [a,b]$.

THEOREM. Let f be continuous on $[a,b]$ and let y lie strictly between $f(a)$ and $f(b)$. Then there exists $c \in (a,b)$ with $f(c) = y$.

PROOF. Set $S = \{x \in [a,b] : f(x) < y\}$. S is non-empty and bounded; let $c = \sup S$. Continuity at c forces $f(c) = y$. $\square$

SECTION 04

Side-by-side · `math compare`

Frequentist vs Bayesian point estimate.

FREQUENTIST

$$\hat{\theta}_{\mathrm{MLE}} = \arg\max_{\theta}\, p(y \mid \theta)$$

Maximises the likelihood — no prior. Uncertainty quantified by the sampling distribution of $\hat{\theta}$ across hypothetical repeats.

BAYESIAN

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta}\, p(\theta \mid y)$$

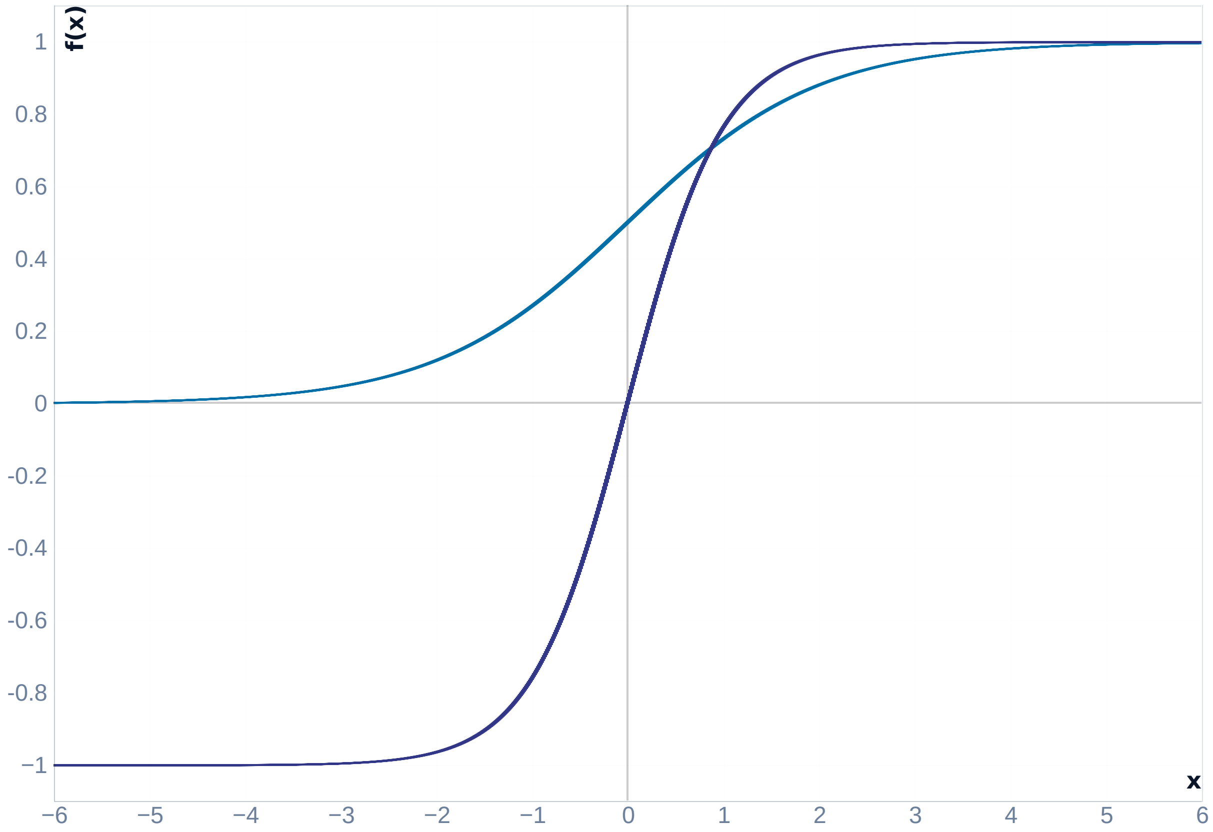
Maximises the posterior — conditions on the prior $p(\theta)$. Uncertainty is the posterior itself, no repeated sampling required.

SECTION 05

Equation + plot · `math canvas`

The sigmoid.

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Maps $\mathbb{R} \to (0,1)$. S-shaped, $\sigma(0) = 0.5$, steepest slope at the origin.

SECTION 06

Matrix · `math matrix` and `math matrix decompose`

The design matrix X.

$$X = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ 1 & x_{21} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}$$

- **SHAPE** — $n \times (p+1)$
- **ROWS** — observations
- **COLS** — intercept + p features
- **RANK** — full-rank for OLS to have a unique solution
- **COLUMN 0** — all-ones, absorbs the intercept

Singular value decomposition.

$$A = U\,\Sigma\,V^\top$$

COMPONENT MATRICES

$U \in \mathbb{R}^{m\times m}$ $\Sigma \in \mathbb{R}^{m\times n}$ $V \in \mathbb{R}^{n\times n}$

SECTION 07

Statistical result · `math stats`

Effect of the treatment.

$$\hat{\beta} = \mathbf{0.42} \pm \mathbf{0.03}$$

95% CI: $[0.36,\ 0.48]$

$p < 0.001 \quad \cdot \quad n = 1{,}204$

For every additional unit of exposure, the outcome rises by 0.42 SD — roughly an ***8%*** *shift on the baseline. Effect size is the headline; the p-value just rules out chance.*

Seven variants. One contract.

math · math derivation · math theorem · math compare · math canvas · math matrix · math stats

KaTeX renders the equations; Lattice provides the room around them.